

CHAPTER – X

10.1 – Conclusion

Plugging into the Future: An Exploration of Electricity Consumption Patterns demonstrates how data analysis and visualization can improve the understanding of electricity usage trends and support smarter energy management. By collecting, analysing, and presenting electricity consumption data through interactive dashboards and analytical tools, the project enables users to identify patterns, monitor performance, and make informed decisions.

The project highlights the importance of efficient energy utilization, demand forecasting, and data-driven planning to support sustainable development. Although challenges such as data quality, maintenance, and prediction limitations exist, the overall system provides valuable insights that can contribute to reducing energy waste and improving operational efficiency.

In the future, the project can be expanded with advanced forecasting models, real-time monitoring, and integration of renewable energy data to create a more intelligent and sustainable electricity management system.

